

AI model used: Claude Sonnet medium

Date of generation: 9/26/2026

Two prompts were used to generate the code.

Prompt 1:

Write me an IPv4 address finder in c++ with no use of external libraries or functions. All processing must be done manually. The requirements are that only numerical digits, periods, and colons are valid input. All other inputs are garbage characters and should be skipped. Port numbers are optional but if included must be valid or else, the input is rejected.

This produced code that largely worked, however, I was not as detailed as I should have been with the user interface. So, I prompted again for an updated user interface and gave the exact function definition from the assignment.

Prompt 2:

Additional requirements. The program should take in one input at a time, check it, then output in this format: Extracted IPv4 address: A.B.C.D (decimal value: N, port: P). This should be the structure for the extraction:

```
// Returns true if a valid address was found, false otherwise.
```

```
// On success: outAddress holds the 32-bit value,
```

```
// and outPort holds the port number, or -1 if no port was present.
```

```
// On failure: outAddress is set to 0 and outPort is set to -1.
```

```
bool extractIPv4(const std::string& str, unsigned long& outAddress, int& outPort);
```

After these two prompts, I only modified the user interface. All other logic and functions are unmodified from the AI-generated output. Additionally, all AI generated comments were removed, and I commented the code to ensure that I understood the output generated.

I verify that I understand all the code that was produced. I tested this code with all test cases from the assignment and additional cases I created (test cases are located in

“test_cases.pdf”). The code produced valid output for all cases I tested. Currently there are no known bugs that I am aware of.